

Optimization of Multivariable Functions

MATH 145 – Applied Calculus

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Optimizing Multivariable Functions

Last lecture, we discussed how to optimize single variable functions.

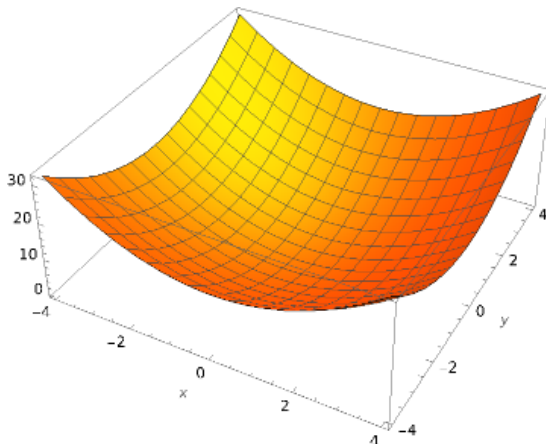
We also discussed how to handle multivariable functions, if they were constrained in a way that we can convert it into a single variable function.

What if we want to optimize a multivariable function across multiple variables?

Note: We are going to discuss three-dimensional functions exclusively this lecture, but the methods we are using can be extended to higher dimensional functions.

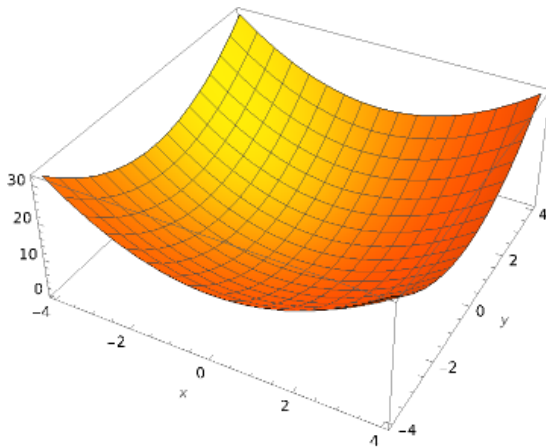
Local Extrema of Functions $f(x, y)$

How can we find the *local extrema* of a function $f(x, y)$ like the graph below?



Local Extrema of Functions $f(x, y)$

Let's try using the *partial derivatives* for $f(x, y)$, $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$, the way we used first derivatives for $f(x)$, $\frac{df}{dx}$.



Critical Points of Functions $f(x, y)$

Critical Points of Functions $f(x, y)$

A **critical point** of a function $f(x, y)$ is a point (x, y) where:

$$\frac{\partial f}{\partial x}(x, y) = \frac{\partial f}{\partial y}(x, y) = 0$$

Example: Critical Points of Functions $f(x, y)$

Example: Find the critical points of the function

$$f(x, y) = x^2 - y^2 + 4x + 2y.$$

Example: Find the critical points of the function

$$g(x, y) = 2xy - 4y + 2x - 3.$$

The Gradient, ∇f

The Gradient

The **gradient** of a function $f(x, y)$ is the coordinate pair $(\partial_x f, \partial_y f)$. We write this as ∇f .

- $\nabla f(x, y)$ is a new function.
- (x_0, y_0) is a critical point of f if $\nabla f(x_0, y_0) = 0$.

Example: The Gradient, ∇f

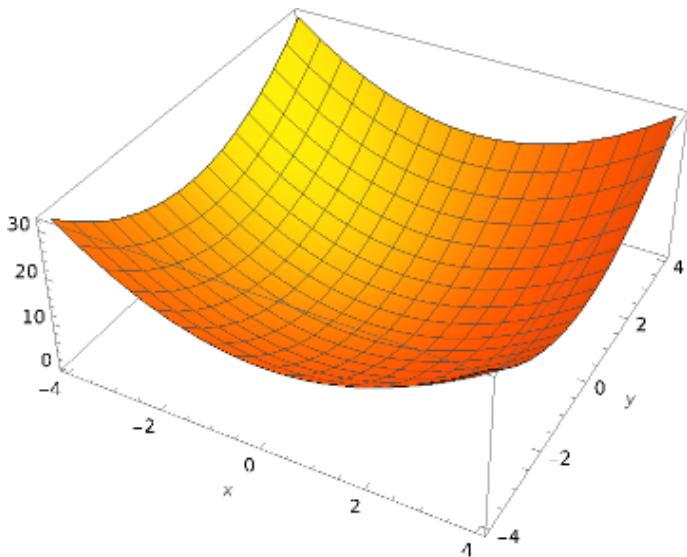
Example: Find the gradient of the function

$$f(x, y) = x^2 - y^2 + 4x + 2y.$$

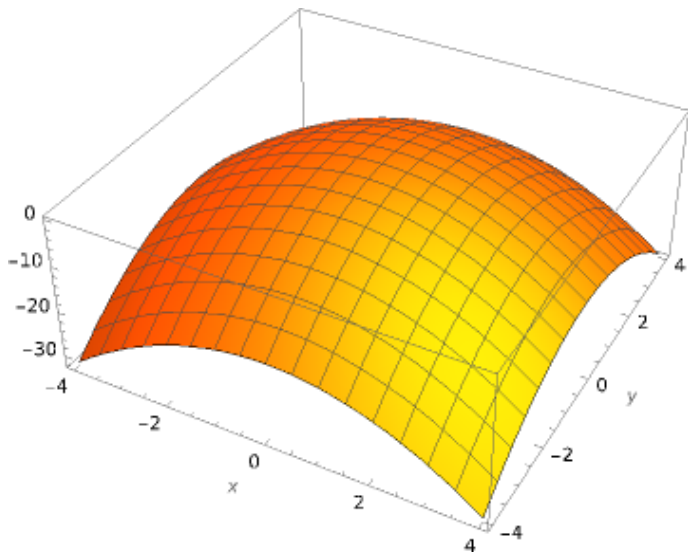
Example: Find the gradient of the function

$$g(x, y) = 2xy - 4y + 2x - 3.$$

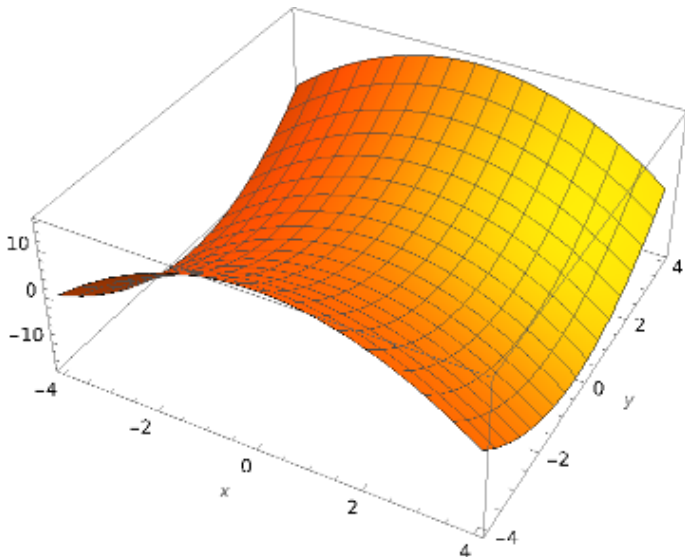
Local Minimum



Local Maximum



Saddle Point



The Second Derivative Test

The Second Derivative Test for Functions of Two Variables

Suppose (x_0, y_0) is a point where $\nabla f(x_0, y_0) = (0, 0)$. Let

$$D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - (f_{xy}(x_0, y_0))^2$$

1. If $D > 0$ and $f_{xx}(x_0, y_0) > 0$, then f has a local minimum at (x_0, y_0) .
2. If $D > 0$ and $f_{xx}(x_0, y_0) < 0$, then f has a local maximum at (x_0, y_0) .
3. If $D < 0$, then f has a saddle point at (x_0, y_0) .
4. If $D = 0$, anything can happen: f can have a local maximum, or a local minimum, or a saddle point, or none of these at (x_0, y_0) .

Example: The Second Derivative Test

Example: Determine the behavior of the function $f(x, y) = x^2 - y^2 + 4x + 2y$ at its critical point.